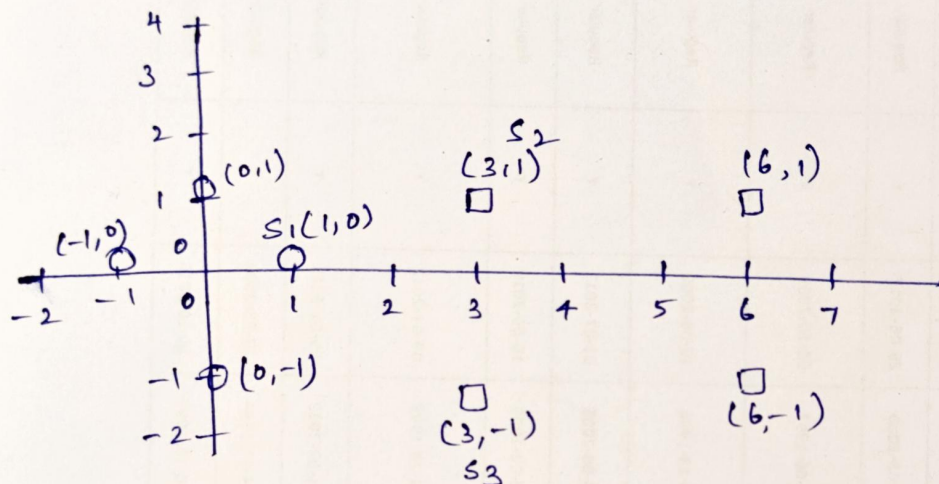


Positively labeled data points

$$\left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$$

Negatively labeled data points

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$$



Support vectors are

$$\left\{ s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \right\}$$

Each vector is augmented with '1' as a bias input

$$s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ after bias } \tilde{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\tilde{s}_2 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

$$s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$\tilde{s}_3 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

Determine w and b to find the hyperplane.

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_1 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_1 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_1 = -1$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_2 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_2 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_2 = +1$$

$$\alpha_1 \tilde{S}_1 \cdot \tilde{S}_3 + \alpha_2 \tilde{S}_2 \cdot \tilde{S}_3 + \alpha_3 \tilde{S}_3 \cdot \tilde{S}_3 = +1$$

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} = +1$$

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = +1$$

$$\alpha_1 (1+0+1) + \alpha_2 (3+0+1) + \alpha_3 (3+0+1) = -1$$

$$\alpha_1 (3+0+1) + \alpha_2 (9+1+1) + \alpha_3 (9-1+1) = 1$$

$$\alpha_1 (3+0+1) + \alpha_2 (9-1+1) + \alpha_3 (9+1+1) = 1$$

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1 \quad \text{--- (1)}$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1 \quad \text{--- (2)}$$

$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1 \quad \text{--- (3)}$$

Solve above three linear equation and find the value of $\alpha_1, \alpha_2, \alpha_3$

Multiply equation (3) ^{by} ~~with~~ -

$$-4\alpha_1 - 9\alpha_2 - 11\alpha_3 = -1 \rightarrow (3)$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1 \rightarrow (2)$$

$$2\alpha_2 - 2\alpha_3 = 0 \rightarrow (4)$$

Multiply equation (1) by -2

$$-4\alpha_1 - 8\alpha_2 - 8\alpha_3 = 2 \rightarrow (1)$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1 \rightarrow (2)$$

$$3\alpha_2 + \alpha_3 = 3 \rightarrow (5)$$

Multiply equation (5) by 2

$$6\alpha_2 + 2\alpha_3 = 6 \rightarrow (5)$$

$$2\alpha_2 - 2\alpha_3 = 0 \rightarrow (4)$$

$$8\alpha_2 = 6$$

$$\alpha_2 = 6/8 = 3/4$$

$$\boxed{\alpha_2 = 0.75}$$

$$\alpha_1 = -3.5$$

$$\alpha_2 = 0.75$$

$$\alpha_3 = 0.75$$

Determine weight

$$\tilde{w} = \sum_i \alpha_i \tilde{S}_i$$

$$= -3.5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$
$$= \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$w = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } b = -2.$$

$x=1, y=0$ line is parallel to y axis
 $x=10, y=1$ line is " to x axis
 $1, 1$ 45° degree x and y axis